

Risk Risk Research Ethics Approval

Project Title

Research and Implementation of the Machine Learning-Based Recommendation Engine to Enhance User Personalization on the E-commerce Platform

Record of Approval

Principal Investigator

I request an ethics peer review and confirm that I have answered all relevant questions in this checklist honestly.	
I confirm that I will carry out the project in the ways described in this checklist. I will immediately suspend research and request new ethical approval if the project subsequently changes the information I have given in this checklist.	
I confirm that I, and all members of my research team (if any), have read and agreed to abide by the Code of Research Ethics issued by the relevant national learned society.	
I confirm that I, and all members of my research team (if any), have read and agreed to abide by the University’s Research Ethics, Governance and Integrity Framework.	
I understand that I cannot begin my research until this ethics application has been approved.	

Name: Soyesh Shrestha

Date:

Student's Supervisor (if applicable)

I have read this checklist and confirm that it covers all the ethical issues raised by this project fully and frankly.I also confirm that these issues have been discussed with the student and will continue to be reviewed in the course of supervision.

Name: Manoj Shrestha

Date:

Reviewer (if applicable)

Date of approval by anonymous reviewer: -

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Project Information

Project Ref	49
Full Name	Soyesh Shrestha
Faculty	Faculty of
Department	School of
Supervisor	Manoj Shrestha
Module Code	ST6047CEM
EFAAF Number	EFAAF
Project Title	Research and Implementation of the Machine Learning-Based Recommendation Engine to Enhance User Personalization on the E-commerce Platform
Date(s)	Date(s)
Created	Created

Project Summary

This research project focuses on the design and implementation of a machine learning-based recommendation engine for the 'Gadgetify' e-commerce platform. The study addresses the problem of "choice overload" and static merchandising by developing a predictive hybrid model capable of analyzing historical user trends to offer personalized product suggestions. Using a desk-based agile research methodology, the project utilizes secondary data to train and evaluate the model's performance against key metrics such as precision, recall, and diversity. A central component of the research is the integration of an ethical framework to mitigate risks associated with algorithmic bias, filter bubbles, and behavioral manipulation. The ultimate goal is to demonstrate how data-driven personalization can enhance user engagement and business profitability while maintaining transparency and ethical integrity.
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Names of Co-Investigators and their organisational affiliation(place of study /employer)	
Is this project externally funded?	No
Are you required to use a Professional Code of Ethical Practice appropriate to your discipline?	Yes
Have you read the Code?	Yes

Project Details

What are the aims and objectives of the project?	The primary aim of this project is to develop a predictive system that assists in e-commerce personalization for the "E-commerce" platform by learning from historical user trends and evaluating the profitability of informed recommendation strategies. The objectives are: To explore and understand the dynamics of user shopping behavior, including key influencing factors such as browsing history and historical trends across the e-commerce platform. To study and evaluate the use of statistical techniques and machine learning methods in predicting user preferences. To examine existing e-commerce personalization and recommendation strategies to identify gaps, limitations, and opportunities for improvement using data-driven approaches. To design and develop a hybrid recommendation model that applies statistical and machine learning techniques on historical e-commerce data. To compare model predictions with the current system's performance to detect potential improvements in user engagement and conversion.
Explain your research design	This design integrates the systematic gathering of secondary data through literature sources, document analysis, and competitor benchmarking with an iterative, cyclical research structure influenced by agile software development principles. Rather than relying on static linear progression, the design uses "sprints." Each cycle begins with identifying a specific technical challenge, followed by data acquisition, model training, and evaluation . Crucially, this design integrates an "ethical review" into every sprint, allowing for the continuous as-

	assessment of risks such as algorithmic bias or filter bubbles as the model evolves .	
Outline the principal methods you will use	The project utilizes the following principal methods: Secondary Data Analysis: Triangulating insights from academic literature, industry whitepapers, and competitor case studies (e.g., Amazon, Netflix) to understand the theoretical and practical landscape of personalization . Data Processing & Feature Engineering: Using Python-based open-source tools (pandas, NumPy) to clean and preprocess large volumes of public interaction data and user demographics. Machine Learning Implementation: Training and validating predictive models using libraries such as scikit-learn to identify patterns in user purchasing habits and product affinities. Simulation & Evaluation: Evaluating model performance by comparing predictions against historical data using metrics like precision, recall, and simulated return on investment to assess business outcomes. Ethical Reflection Cycles: Conducting structured reviews at the end of each development cycle to interpret results in light of ethical principles like fairness and transparency.	
Are you proposing to use a validated scale or published research method / tool?		Yes
Does your research seek to understand, identify, analyse and/or report on information on terrorism or from terrorist organisations, require access to terrorist groups or those convicted of terrorist offences or relate to terrorism policies in other international jurisdictions?		No
		No

Does your research seek to understand, identify, analyse and/or report on information for other activities considered illegal in the UK and/or in the country you are researching in?	
Are you dealing with Secondary Data? (e.g. sourcing info from websites, historical documents)	Yes
Is this data publicly available?	Yes
Could an individual be identified from the data? e.g. identifiable datasets where the data has not been anonymised or there is risk of re-identifying an individual	No
Are you dealing with Primary Data involving people? (e.g. interviews, questionnaires, observations)	No

Are you dealing with personal data?		Yes
Please specify what personal data you will be collecting.	The research utilizes secondary, anonymized data sourced from publicly accessible open datasets rather than primary collection. The specific data points processed include: User Demographics: Non-identifiable attributes such as age or location. Behavioral Data: Historical interaction logs, including product views, search queries, and purchase histories.	
Are you dealing with sensitive data (special category data)?		No
Will the Personal or Sensitive data be shared with a third party?		No
Will the Personal or Sensitive data be shared outside of the European Economic Area(EEA)?		No
Is the project solely desk based? (e.g. involving no laboratory, workshop or offcampus work or other activities which pose significant risks to researchers or participants)		Yes
Will the data collection, recruitment materials or any other project documents be in any language other than English?		No
Are there any other ethical issues or risks of harm raised by the study that have not been covered by previous questions?		No

DBS (Disclosure & Barring Service) formerly CRB (Criminal Records Bureau)

Question		Yes	No
Does the study require DBS (Disclosure & Barring Service) checks?			×
If YES, Please give details of the level of check, serial number, date obtained and expiry date (if applicable)			
If NO, does the study involve direct contact by any member of the research team with children or young people under 18 years of age?			×
If NO, does the study involve direct contact by any member of the research team with adults who have learning difficulties, brain injury, dementia, degenerative neurological disorders?		×	×
If NO, does the study involve direct contact by any member of the research team with adults who are frail or physically disabled?			×
If NO, does the study involve direct contact by any member of the research team with adults who are living in residential care, social care, nursing homes, re - ablement centres, hospitals or hospices ?			×
If NO, does the study involve direct contact by any member of the research team with adults who are in prison, remanded on bail or in custody?			×
If you have answered YES to any of the questions above please explain the nature of that contact and what you will be doing			

External Ethics Review

Question		Yes	No
Will this study be submitted for ethical review to an external organisation ? (e.g.Another University, Social Care, National Health Service, Ministry of Defence, Police Service and Probation Office)			×
If YES, name of external organisation			
Will this study be reviewed using the IRAS system?			×
Has this study previously been reviewed by an external organisation?		×	×

Confidentiality, security and retention of research data

Question		Yes	No
What data are you collecting / using / recording?	I am using secondary, anonymized e-commerce interaction datasets (e.g., clickstream logs, purchase history, product ratings) sourced from public open-source repositories (e.g., Kaggle, UCI Machine Learning Repository). No PII (Personally Identifiable Information) such as names, addresses, or phone numbers will be processed.		
Are there any reasons why you cannot guarantee the full security and confidentiality of any personal or confidential data collected for the study?			×
Please provide an explanation			
Is there a significant possibility that any of your participants, and associated persons, could be directly or indirectly identified in the outputs or findings from this study?			×
Please provide an explanation		The datasets used are secondary and already anonymized (stripped of personal identifiers) before acquisition.	
Is there a significant possibility that a specific organisation or agency or participants could have confidential information identified, as a result of the way you write up the results of the study?			×
Please provide an explanation		The data sources are public. No private corporate data or trade secrets are being accessed.	
Will any members of the research team retain any personal or confidential data at the end of the project, other than in fully anonymised form?			×
Please provide an explanation		All datasets and derived models will be deleted from local storage upon the successful completion and grading of the degree.	
Will you or any member of the team intend to make use of any confidential information, knowledge, trade secrets obtained for any other purpose than the research project ?			×
Please give an explanation			
Have you taken necessary precautions for secure data management, in accordance with data protection and CU Policy		×	
Specify location (physical and electronic) where data will be stored	Data will be stored on a password-protected local machine. Code and non-sensitive documentation will be stored in a private GitHub repository.		
Will you be responsible for destroying the data after study completion?		×	
If NO, who will be responsible for this?			
Please explain how any identifiable and anonymous data will be destroyed		Permanent deletion of files from the local hard drive and removal of any temporary cloud storage buckets used for model training.	
Planned disposal date		after graduation	

Participant Information and Informed Consent

Question		Yes	No
Will all the participants be fully informed BEFORE the project begins why the study is being conducted and what their participation will involve ?			×
Please explain why	This is a desk-based study using secondary public data. There are no human participants to recruit, inform, or debrief.		
Will every participant be asked to give written consent to participating in the study, before it begins ?			×
If NO, please explain how you will get consent from your participants.If not written consent, explain how you will record consent	This is a desk-based study using secondary public data. There are no human participants to recruit, inform, or debrief.		
Will all participants be fully informed about what data will be collected, and what will be done with this data during and after the study ?			×
If NO, please specify	This is a desk-based study using secondary public data. There are no human participants to recruit, inform, or debrief.		
Please explain what recordings (audio, visual or both) will be made and how you will gain consent for recording participants	This is a desk-based study using secondary public data. There are no human participants to recruit, inform, or debrief.		
Will all participants understand that they have the right not to take part at any time, and/or withdraw themselves and their data from the study if they wish?			×
If NO, please explain why	This is a desk-based study using secondary public data. There are no human participants to recruit, inform, or debrief.		
Will every participant understand that there will be no reasons required or repercussions if they withdraw or remove their data from the study?			×
If NO, please explain why	This is a desk-based study using secondary public data. There are no human participants to recruit, inform, or debrief.		
Does the study involve deceiving, or covert observation of, participants?			×
Will you debrief them at the earliest possible opportunity?			×
If NO to debrief them, please explain why this is necessary	This is a desk-based study using secondary public data. There are no human participants to recruit, inform, or debrief.		

Risk of harm, potential harm and disclosure of harm

Question		Yes	No
Is there any significant risk that the study may lead to physical harm to participants or researchers ?			×
If you have answered Yes, please explain how you will take steps to reduce or address those risks. If you have answered No, explain why you believe this is the case	The research is computational and conducted entirely on a computer.		
Is there any risk that your study may lead or result in harm to the reputation of the University Group, its researchers or the organisations involved in the study?			×
If you have answered Yes, please explain how you will take steps to reduce or address those risks. If you have answered No, explain why you believe this is the case	The research adheres to standard data science practices and ethical guidelines regarding AI transparency.		
Is there a risk that the study will lead to participants to disclose evidence of previous criminal offences, or their intention to commit criminal offences?			×
If you have answered Yes, please explain how you will take steps to reduce or address those risks. If you have answered No, explain why you believe this is the case	The research adheres to standard data science practices and ethical guidelines regarding AI transparency.		
Is there a risk that the study will lead participants to disclose evidence that children or vulnerable adults are being harmed, or at risk or harm?			×
If you have answered Yes, please explain how you will take steps to reduce or address those risks. If you have answered No, explain why you believe this is the case	The research adheres to standard data science practices and ethical guidelines regarding AI transparency.		
Is there a risk that the study will lead participants to disclose evidence of serious risk of other types of harm ?			×
If you have answered Yes, please explain how you will take steps to reduce or address those risks. If you have answered No, explain why you believe this is the case	The research adheres to standard data science practices and ethical guidelines regarding AI transparency.		
Will participants be made aware of the circumstances in which disclosure has implications for confidentiality?			×

Payments to participants

Question		Yes	No
Do you intend to offer participants cash payments or any kind of inducements, or reward for taking part in your study ?			×
If YES, please explain what kind of payment you will be offering(e.g.prize draw or store vouchers)			
Is there any possibility that such payments or inducements will cause participants to consent to risks that they might not otherwise find acceptable ?			
If YES, please explain)			
Is there any possibility that the prospect of payment or inducements will influence the data provided by participants in any way ?			
If YES, please explain)			
Will you inform participants that accepting payments or inducements does not affect their right to withdraw from the study at any time ?			

Capacity to give valid consent

Question		Yes	No
Do you propose to recruit any participants?			
Do you propose to recruit any participants who are children or young people under 18 years of age?			×
Do you propose to recruit any participants who are adults who have learning difficulties, mental health conditions, brain injury, advanced dementia, degenerative neurological disorders ?			×
Do you propose to recruit any participants who are adults who are physically disabled and cannot provide written and/or verbal consent			×
Do you propose to recruit any participants who are with adults who are living in residential care, social care, nursing homes, reablement centres, hospitals or hospices ?			×
Do you propose to recruit any participants who are with adults who are in prison, remanded on bail or in custody?			×
If you have answered YES to any of the questions above please explain overcome any challenges to gaining valid consent			
Do you propose to recruit any participants with possible communication difficulties, including difficulties arising from limited use of knowledge of the English language ?			×
If YES, please explain how you will overcome any challenges to gaining valid consent			
Do you propose to recruit participants who may not be able to fully understand the nature of the study, the foreseen implications or cannot provide consent?			×
If YES, please explain how you will overcome any challenges to gaining valid consent			

Recruiting Participants

Question		Yes	No
Who are the participants?			
How are participants being recruited? Please provide details on all methods of recruitment you intend to use			
Do you foresee any conflict of interest?			X
Please explain how will this conflict of interest be addressed			

Online and Internet Research

Question		Yes	No
Will any part of your project involve collecting data via the internet or social media?		X	
If YES, please explain how you will obtain permission to collect data by these means	The project involves downloading pre-existing, open-source datasets from internet repositories (e.g GitHub) for training the machine learning model.		
Will this require consent to access?			X
If NO, please explain how you will get permission/ consent' to collect this information?	The datasets are publicly available under open data licenses (e.g., Creative Commons, MIT License) which explicitly permit research use without requiring individual user consent.		
Will you be collecting data using an online questionnaire/ survey tool? (e.g. BoS, Filemaker)?			
If YES, please explain which software and how you are ensuring appropriate data security			
Is there a possibility that the study will encourage children under 18 to access inappropriate websites, or correspond with people who pose risk of harm ?			X
If YES, please explain further			
Will the study incur any other risks that arise specifically from the use of electronic media ?			X
If YES, please explain further			

Information gathered from human participants

Question		Yes	No
Primary			
Does your project involve primary data collection from human participants via questionnaires, focus groups, interviews, psychological tests, photography/videography etc.?			X
If YES, Please detail the information to be collected and methods that will be used.			
Is there the possibility of physical or psychological harm to the researcher(s) or the participants?			X
If YES, please explain the possible harm and action taken to reduce/remove the risk			
Are any specific exclusions needed to prevent possible harm to participants (e.g. excluding people with known mental health problems)?			X
If YES, please explain exclusions needed and how these will be carried out			
Are any of the questionnaires or other tests being used in the research diagnostic for specific clinical conditions?			X
If YES, Please explain how you will take steps to reduce or address these risks			